

Data Flow Diagram

Date	14 July 2026
Project Name	Strategic Product Placement Analysis
Maximum Marks	2 Marks

Data Flow Diagram

The Data Flow Diagram (DFD) illustrates how retail data flows through the Strategic Product Placement Analysis system. The process begins with collecting the retail dataset, followed by data preparation, visualization in Tableau, dashboard creation, story generation, and web integration using Flask.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person or organization outside the system that provides data or receives analysis results (e.g., Retail Manager, Marketing Analyst).
Rectangle with numbered header	Process	An activity that transforms incoming data into meaningful information (e.g., Data Collection, Data Visualization).
Rectangle (solid fill, no number)	Data Store	Stores the retail dataset and processed information for visualization and analysis.

Labeled arrow	Data Flow	Represents the movement of data between entities, processes, and data stores.
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External Entity

Retail Manager / Marketing Analyst

Process 1 – Data Collection

- Collect the retail sales dataset.
 - Verify the completeness and accuracy of the data.
 - Import the dataset into Tableau Public.
 - Prepare the dataset for further processing.
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Process 2 – Data Storage & Processing

- Store the dataset in CSV format.
 - Clean missing and inconsistent values.
 - Transform data into the required format.
 - Create calculated fields such as:
 - Price Difference
 - Sales per Price
 - Organize data for visualization.
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Data Store

Retail Product Dataset Contains:

Contains

- Product Number

- Product Category
 - Product Position
 - Price
 - Competitor Price
 - Promotion
 - Seasonal Indicator
 - Consumer Demographics
 - Units Sold
 - Sales Volume
 - Calculated Fields
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Process 3 – Data Visualization

- Connect the dataset with Tableau Public.
 - Create interactive charts and graphs.
 - Develop Tableau Dashboard.
 - Create Tableau Story.
 - Apply filters and calculated fields.
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Process 4 – Web Application

- Integrate Tableau Dashboard using Flask.
 - Display dashboards and stories through the web interface.
 - Provide interactive filtering and navigation.
 - Enable users to explore retail business insights.
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Output

- Product Category Analysis
- Product Position Analysis
- Consumer Demographics Analysis
- Price vs Competitor Price Analysis
- Promotion Effectiveness Analysis
- Seasonal Product Analysis
- Sales Performance Dashboard
- Interactive Tableau Dashboard
- Tableau Story
- Business Insights for Strategic Product Placement.

Example

The diagram below illustrates how the symbols above are combined to form the complete Data Flow Diagram for the **Strategic Product Placement Analysis** system.

